ZEAL COLLEGE OF ENGINEERING & RESEARCH, PUNE
PRESENTS

BRIDGE NOVA

Offical Event Rulebook

AVISHKAR '26
WHERE CURIOSITY BECOMES CREATION

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1. Introduction

- RoboRush is a robotics competition where teams design robots to navigate an obstacle-filled track without human intervention.
- The event evaluates obstacle detection and avoidance, path planning, decision-making, robot stability, speed, and sensor integration.

2. Objective

- The goal is to build a robot that navigates from start to end while efficiently avoiding obstacles in the minimum amount of time.
- Victory is awarded to the robot with the fastest and cleanest run.

3. Eligibility of Participation

- Only team participation is allowed, consisting of 2-4 members per team.
- Participation is restricted to undergraduate (UG) students.
- Each team is limited to only one robot.

4. Robot Specification

- Robots must be remote-controlled with an onboard power supply and processor.
- Size must not exceed 30 cm x 30 cm (L x W), with a preferred height under 30 cm.
- The maximum weight allowed is 1-1.5 kg.
- Robots are prohibited from jumping, flying, climbing, damaging obstacles, marking the arena, or leaving parts behind; violations result in immediate disqualification.

5. Arena & Track

- The track layout is revealed 30 minutes prior to the competition.
- The arena may include static and dynamic obstacles, narrow paths, dead ends, turns, and checkpoints.
- Surface conditions, such as friction, lighting, and colors, may vary and are not guaranteed to be consistent.
- Robots must be adaptable, as pre-fed maps are not permitted.

6. Competition Format

- Round 1 (Basic Level): Focuses on navigation and stability on a simple track with four checkpoints (P1-P4). Teams have a maximum of 15 minutes and 3 runs, with the best run considered.
- Round 2 (Advanced Level): Features a complex track with advanced obstacles like rotating discs, flap bridges, mystery obstacles, and hidden paths. Teams are allowed 3 runs with up to 5 checkpoints.

7. Game Rules

- Timing starts when the robot crosses the start line and ends at the finish line.
- Participants are allowed to adjust sensors, change switch positions, and repair the robot during their allotted time.
- It is strictly forbidden to touch the robot with hands or feed it track data after the layout reveal.

8. Attempts & Restarts

- Teams are allowed a maximum of 3 restarts from the previous valid position.
- If a robot becomes stuck, it must be corrected within these attempts or the run will be aborted.

9. Judging Criteria

- Points are awarded for: Each checkpoint (+10), lap completion (+10), shortest path (+5), no collisions (+5), fastest time (+5), and best design (+5).
- 10 points are deducted for any damage to the arena.

10. Tie-Breaker

- Ties are resolved based on the number of checkpoints, the fastest final run, and having fewer penalties.

11. Technical Inspection

- A mandatory inspection is conducted before the competition to ensure robots meet all specifications.
- Failure to meet specifications results in disqualification.

12. General Rules

- The judges' decisions are final, and teams must report any issues directly to them.
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- Organizers reserve the right to modify rules if necessary.